

Exp 4: Data modelling and form design for a business application in Zoho Creator.

1. Aim

The objective of this experiment is to design a normalized relational data model and construct a comprehensive low-code business application form—the **Vehicle Service & Repair Request Management System**—in Zoho Creator. This includes defining entity schemas, field data types, data validation rules, mandatory field constraints, and creating associated report views for enterprise record management.

2. Procedure

The development of the Vehicle Service & Repair Request Management System followed a structured low-code engineering methodology, comprising relational data modelling, schema specification, form field construction, property configuration, and report view deployment.

2.1 Relational Data Modelling & ER Diagram

To ensure high data integrity, eliminate data redundancy, and support multi-table relational links, an Entity-Relationship (ER) model was constructed. The architecture comprises four core entities: CUSTOMER, VEHICLE, SERVICE_REQUEST, and TECHNICIAN.

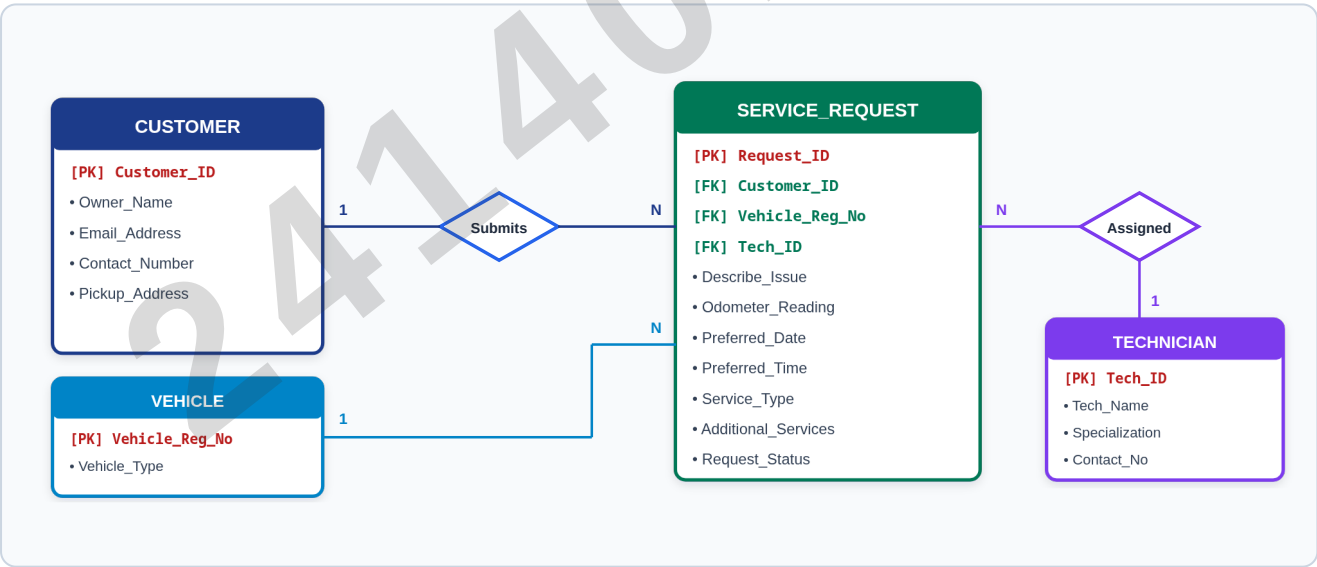


Figure 1: Professional Entity-Relationship (ER) Schema Diagram for Vehicle Service & Repair Management System

2.2 Data Dictionary & Schema Specification

Table 1 details the normalized data dictionary, mapping logical attributes to physical Zoho Creator field types, field link names, validation constraints, and business rules configured for the primary Service_Request_Form.

Table 1: Normalized Data Dictionary for Service_Request_Form Schema

Field Name	Field Link Name	Data Type	Constraints & Validation Rules	Business Purpose & Description
Owner Name	Owner_Name	Name (Composite)	Mandatory (First/Last Name required)	Full legal name of the vehicle owner submitting the request.
Email Address	Email_Address	Email	Mandatory, Standard RFC 5322 Email Validation	Primary electronic contact for automated status updates and digital invoices.
Pickup Address	Pickup_Address	Address (Composite)	Mandatory (Line 1, City, State, Pincode)	Physical address for vehicle pickup and doorstep delivery coordination.
Contact Number	Contact_Number	Phone	Mandatory, Numeric Mobile Validation	Direct telephone contact for service advisor and driver communication.
Vehicle Registration Number	Vehicle_Registration_Number	Single Line	Mandatory, Uppercase Alphanumeric	Unique government registration license plate number of the vehicle.
Describe the Issue	Describe_the_Issue	Multi Line	Optional, Plain Text Box	Detailed customer statement of mechanical symptoms, noise, or requested repairs.
Odometer Reading (km)	Odometer_Reading_km	Number	Optional, Positive Integer (≥ 0)	Current mileage distance recorded on vehicle dashboard at check-in.
Preferred Service Date	Preferred_Service_Date	Date	Mandatory, Date $\geq$ Current Date	Requested calendar date for booking service bay and technician allocation.
Preferred Drop-off Time	Preferred_Drop_off_Time	Time	Optional, Standard Time Picker	Target time slot for delivering the vehicle to the service workshop.
Vehicle Type	Vehicle_Type	Drop Down	Options: Hatchback,	Classification of vehicle chassis type

Field Name	Field Link Name	Data Type	Constraints & Validation Rules	Business Purpose & Description
			Sedan, SUV, EV, Commercial	for workshop bay assignment.
Service Type	Service_Type	Drop Down	Options: General Service, Periodic Repair, Inspection	Primary maintenance tier requested by customer.
Additional Services Requested	Additional_Services_Requested	Multi Select	Options: Wheel Alignment, Foam Wash, Ceramic Coating	Add-on specialized maintenance items selected by owner.

2.3 Step-by-Step Implementation Procedure

- Application Instantiation:** Logged into the Zoho Creator developer console and created a new cloud application titled *"Vehicle Service & Repair Request Management"*.
- Form Builder Canvas Setup:** Navigated to **DEVELOP > Forms** and created a new blank form titled `Service_Request_Form`.
- Drag-and-Drop Field Assembly:** Dragged composite Name, Email, Address, Phone, Number, Single Line, Multi Line, Date, Time, Drop Down, and Multi Select fields from the component palette onto the form layout canvas.
- Field Property Configuration:** Set exact field labels, field link names, mandatory validation flags, placeholder text, and option lists in the right-hand Field Properties inspector panel.
- Report View Deployment:** Configured and deployed an automatic list report view titled `Service_Request_Form Report` to provide multi-column data filtering, record inspection, and administrative management.

3. Output

The following figures illustrate the deployed form builder workspace, live runtime user form, and record management report view in Zoho Creator:

Figure 2: Zoho Creator Form Builder Canvas & Field Properties Inspector for Service_Request_Form

Figure 3: Live Application Runtime & User Input Form for Vehicle Service & Repair Requests

The screenshot displays the 'Service_Request_Form Report' view in Zoho Creator. The sidebar on the left contains navigation links for 'Vehicle Service & Repair Request...', 'Service_Request_Form', and 'Service_Request_Form Report'. The main area shows a table of customer records with columns for Owner Name, Email Address, and Pickup Address. A detailed form view is open for the first record, showing fields for Owner Name, Email Address, Pickup Address, Contact Number, Vehicle Registration Number, Describe the Issue, Odometer Reading (km), Preferred Service Date, Preferred Drop-off Time, Vehicle Type, Service Type, Additional Services Requested, and Authorization.

Owner Name	Email Address	Pickup Address
Pravinraj P	pravinraj.service@example.com	123 Innovation Way, Tech Park, OMR Road, Karapakkam, Chennai, Tamil Nadu, 600097, India

Owner Name	Pravinraj P
Email Address	pravinraj.service@example.com
Pickup Address	123 Innovation Way, Tech Park, OMR Road, Karapakkam, Chennai, Tamil Nadu, 600097, India
Contact Number	+917010099999
Vehicle Registration Number	TN-01-AB-1234
Describe the Issue	Scheduled 30,000 km periodic service check. Please inspect brake pads and top up engine fluid.
Odometer Reading (km)	30150
Preferred Service Date	18-Aug-2026
Preferred Drop-off Time	10:00:00
Vehicle Type	Hatchback
Service Type	Routine Service
Additional Services Requested	Car Wash Interior Cleaning Tyre Rotation
Authorization	I authorize the service center to prepare an estimate and proceed with the service/repair

Figure 4: Service_Request_Form Report View Displaying Submitted Customer Records

4. Result

The normalized relational data model, ER schema, data dictionary, user interface form, and administrative report view for the **Vehicle Service & Repair Request Management System** were successfully designed, configured, deployed, and verified in Zoho Creator in accordance with the experiment requirements.